

REVA Academy for Corporate Excellence (RACE)

Machine Learning and Explainability

Mentor: Ratnakar Pandey

Syllabus, Lesson Plan and Evaluation

Syllabus and Resources			
Course Description	This course introduces core machine learning concepts and practical model building techniques, covering classification, regression, clustering, and dimensionality reduction using industry-standard tools. It emphasizes model evaluation, interpretability, and responsible AI practices. Learners will gain hands-on experience in explaining model decisions using techniques such as SHAP and LIME, enabling them to build transparent, trustworthy, and ethically responsible machine learning solutions for real-world applications.		
Course Objectives	By the end of this course, learners will be able to: 1. Understand fundamental machine learning concepts, including supervised, unsupervised, and reinforcement learning. 2. Apply classification, regression, and clustering algorithms to solve real-world data problems. 3. Evaluate and compare machine learning models using appropriate performance metrics and validation techniques. 4. Implement end-to-end machine learning workflows using libraries such as scikit-learn and gradient boosting frameworks. 5. Interpret and explain model predictions using explainable AI techniques like SHAP, LIME, and feature importance. 6. Apply responsible AI principles by identifying bias, ensuring fairness, and validating model decisions.		
CO#	Course Outcomes	POs	PSOs
C01	Understand core machine learning concepts including supervised and unsupervised learning.	1, 2, 5	1, 2
C02	Apply classification, regression, and clustering algorithms on structured datasets.	1, 2	4
C03	Evaluate model performance using appropriate metrics and validation techniques.	1, 5	1, 2
C04	Utilize libraries like scikit-learn and XGBoost for real-world ML applications.	3, 4	3
C05	Apply model interpretability techniques (SHAP, LIME, feature importance).	4, 5	3, 4
C06	Demonstrate responsible AI practices by interpreting and validating model decisions.	4, 5	3, 4

For more information, visit the program LMS <https://racelms.reva.edu.in/dashboard>

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Legend: LTP: Lecture:1 Tutorial:0 Practical:3

CIE – Continuous Internal Examination

MEE – Module End Exam

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Course Contents	Unit 1: Introduction to Machine Learning: Types of learning – supervised, unsupervised, reinforcement learning. ML pipeline, dataset preparation, feature selection. Linear models – Linear Regression, Logistic Regression. Unit 2: Classification and Regression Models: Decision Trees, Random Forests, k-NN, SVM, Naive Bayes. Gradient Boosting methods (XGBoost, LightGBM). Model evaluation metrics – accuracy, precision, recall, F1-score, ROC-AUC. Unit 3: Clustering and Dimensionality Reduction: K-means, Hierarchical Clustering, DBSCAN. Dimensionality reduction using PCA, t-SNE. Applications in exploratory analysis and pattern discovery. Unit 4: Model Explainability and Responsible AI: Introduction to explainable AI (XAI). Global vs local interpretability. SHAP values, LIME, Permutation Importance. Bias detection and fairness metrics. Interpreting black-box models.
Suggested Labs	Lab 1: Build a Linear regression with necessary data cleaning, feature engineering and evaluate performance using scikit-learn Lab 2: Build a logistic regression classifier with necessary data cleaning, feature engineering and evaluate performance using scikit-learn Lab 3: Implement decision trees and random forests; visualize feature importance using several classifier techniques and build a leader board. Lab 4: Perform k-means clustering on customer segmentation dataset and plot clusters. Lab 5: Apply PCA for dimensionality reduction and visualize transformed data. Lab 6: Use SHAP and LIME for interpreting predictions of tree-based models. Lab 7: Mini project – Train a classification model and present explainability analysis. Lab 8: Build a time series model using techniques such as Holt Winters and Prophet
Tools/Softwares	Python 3.xxx, Google Colab, Jupyter notebook
Books and Resources	Foundational Machine Learning Books - Pattern Recognition and Machine Learning – Christopher M. Bishop - The Elements of Statistical Learning – Hastie, Tibshirani & Friedman.

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	- Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow – Aurélien Géron Explainability & Responsible AI - Interpretable Machine Learning – Christoph Molnar - Explainable AI – Samek, Montavon, Vedaldi et al. Online Resources & Libraries - scikit-learn Documentation & User Guide – Practical implementation and examples for ML pipelines. - SHAP Documentation – Authoritative resource for Shapley value-based explanations. - LIME GitHub & Tutorials – Local interpretability for black-box models. - Google Responsible AI & IBM AI Fairness 360 – Toolkits and guidelines for ethical and fair AI.
Mentor's Profile	https://www.linkedin.com/in/ratnakarpandey/

Lesson Plan		
Day #	Theory	Practice/Lab/Projects
1.	Unit 1: Introduction to Machine Learning: Types of learning – supervised, unsupervised, reinforcement learning. ML pipeline, dataset preparation, feature selection. Linear models – Linear Regression, Logistic Regression.	Lab 1: Build a Linear regression with necessary data cleaning, feature engineering and evaluate performance using scikit-learn Lab 2: Build a logistic regression classifier with necessary data cleaning, feature engineering and evaluate performance using scikit-learn.
2.	Unit 2: Classification and Regression Models: Decision Trees, Random Forests, k-NN, SVM, Naive Bayes. Gradient Boosting methods (XGBoost, LightGBM). Model evaluation metrics – accuracy, precision, recall, F1-score, ROC-AUC.	Lab 3: Implement decision trees and random forests; visualize feature importance using several classifier techniques and build a leader board.
3.	Unit 3: Clustering and Dimensionality Reduction: K-means, Hierarchical Clustering, DBSCAN. Dimensionality reduction using PCA, t-SNE. Applications in exploratory analysis and pattern discovery.	Lab 4: Perform k-means clustering on customer segmentation dataset and plot clusters. Lab 5: Apply PCA for dimensionality reduction and visualize transformed data.

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4.	Unit 4: Model Explainability and Responsible AI: Introduction to explainable AI (XAI). Global vs local interpretability. SHAP values, LIME, Permutation Importance. Bias detection and fairness metrics. Interpreting black-box models.	Lab 6: Use SHAP and LIME for interpreting predictions of tree-based models. Lab 7: Mini project – Train a classification model and present explainability analysis.
5.	Timeseries Modeling + Revision	Lab 8: Build a time series model using techniques such as Holt Winters and Prophet
6.	Revision and Final Exam.	

Evaluation Components (EC)				
No	Name	Type	Weight	Date
CIE1	Attendance and Class Participation/LMS	Open	10%	-
CIE2	Quiz (MCQ)	Open	20%	
CIE3	Team assignment	Open	20%	TBD
MEE	Module End Exam	Open Book/Lab	50%	Day 6

Instructions:

- Active participation is expected.
- Pre-readings, datasets and scripts will be shared through LMS.
- Lab Manuals will be shared through LMS.

Annexure 1:

Bloom's Taxonomy Level

CO#	Remember (L1)	Understand (L2)	Apply (L3)	Analyse (L4)	Evaluate (L5)	Create (L6)
CO1				✓		
CO2				✓		
CO3			✓			
CO4			✓			
CO5			✓			
CO6			✓			

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Annexure 2: Course Articulation Matrix

CO#/POs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	M	H	H	H	H	L	L	L	M	L	L	H	H	M
CO2	M	H	H	H	H	L	L	L	M	L	L	H	H	M
CO3	M	H	H	H	H	L	L	L	M	L	L	H	H	M
CO4	M	M	H	H	H	L	L	M	M	L	L	H	H	M
CO5	M	H	H	H	H	L	L	M	M	L	L	H	H	M
CO6	M	H	H	H	H	L	L	M	M	L	L	H	H	H

Legend: Low:1; Medium:2, High:3

Annexure 3: MTech/MSc AI Program Outcomes (POs)

The Semester-Modular MTech/M. Sc. Artificial Intelligence degree Programme is designed to achieve the following objectives. After successful completion of the Programme, the graduates shall be able to:

Sl. No	Program Outcomes (POs)
PO1	Demonstrate in-depth knowledge in Artificial Intelligence and related disciplines, including global perspectives, with the ability to analyze, synthesize, and integrate existing and emerging knowledge for technological advancement.
PO2	Identify, define, and critically analyze complex problems across organizational and technological domains. Apply independent judgment and research methodologies to derive innovative and context-sensitive solutions.
PO3	Design and implement AI-based systems by evaluating multiple solution paths, considering technical feasibility, economic viability, environmental sustainability, public health and safety, and ethical implications.
PO4	Explore unfamiliar problems through systematic literature review, experimentation, and the application of advanced research techniques, leading to the development of novel insights and innovative solutions.
PO5	Select, adapt, and apply state-of-the-art tools, technologies, and AI frameworks for modeling, simulation, analysis, and deployment of intelligent systems, understanding their strengths and limitations.
PO6	Work effectively as an individual and in diverse teams, demonstrating leadership, self-management, and collaborative skills in multidisciplinary and multicultural environments.
PO7	Apply engineering and management principles to plan, execute, and evaluate AI-based projects, while considering economic, financial, and operational constraints in real-world settings.
PO8	Communicate ideas, technical concepts, and research findings clearly and effectively through well-structured documentation, presentations, and discussions with technical and non-technical audiences.
PO9	Recognize the importance of continuous learning and develop the ability to independently pursue upskilling, certifications, research, and knowledge enhancement in the rapidly evolving field of Artificial Intelligence.
PO10	Demonstrate ethical behavior, intellectual honesty, and integrity in research and professional practices, considering the societal and environmental impact of AI systems and contributing to sustainable development.

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PO11	Critically evaluate outcomes of personal and professional actions, learn from experiences and failures, and make necessary improvements without relying solely on external feedback.
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Annexure 4: MTech/MSc AI Programme Specific Outcomes (PSOs)

After successful completion of the Programme, the graduates shall be able to:

Sl. No	Program Specific Outcomes (POs)
PSO1	Graduates will be able to design and develop intelligent systems using core AI techniques such as machine learning, deep learning, natural language processing, and computer vision to solve real-world multidisciplinary problems.
PSO2	Graduates will demonstrate the ability to build, deploy, and manage AI models in production environments using modern cloud platforms and MLOps tools, integrating industry-recognized certifications.
PSO3	Graduates will possess the skills to identify research problems, conduct systematic investigations, and contribute to the body of knowledge in AI through capstone projects, scholarly publications, and presentations aligned with ethical research practices.
PSO4	Graduates will be prepared for successful careers in industry, academia, or entrepreneurship with the ability to work in teams, engage in lifelong learning, and apply AI solutions responsibly, adhering to ethical and societal norms.